

Assessing hurricane exposure for epidemiological research

Focus on exposure to tropical cyclone-related winds

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Overview

While there are many hazards associated with tropical cyclones, in this session we'll focus on the hazard of **high winds**.

We'll discuss how wind intensity can be measured for tropical cyclones and how this can be leveraged for **exposure assessment** for epidemiology studies.

The principles apply with a variety of available datasets and wind field models. I will show examples from one set of R packages that my research group developed and made available.

We'll start by talking about central measures of storm intensity, focusing on **storm track data**.

We'll then move into a discussion of **wind field modeling** and how that has advantages in terms of estimating winds local to each study subject. We'll discuss how wind field models can be used for **exposure assessment** for epidemiology. We'll talk some about binary versus continuous exposure measures.

We'll end by discussing some **alternatives to wind field modeling**, as well as briefly discuss how exposure assessment that is based on wind might differ from exposure assessment based on **other hazards** of tropical cyclones.

Many of the concepts in this session are covered in the following article. I will make this available to all the participants, for anyone who would like to go deeper based on the overview I give today.

Anderson, Schumacher, and Done. 2022. Exposure Assessment for Tropical Cyclone Epidemiology. *Current Environmental Health Reports*. 9:104–119.

We are at the end of a few hours of talks. I have focused on sharing the information for this session in a way that is visual and hopefully engaging.

For full citations and deeper dives into specific concepts that I'll cover, please take a look at the paper I'll be sharing, as well as vignettes for the R packages I'll use as examples.

Storm tracks data

Tropical cyclones as organized systems

Tropical cyclones are **organized** weather systems. As a result they can—and are—tracked and measured **as a system**.

Tropical cyclones as organized systems

In this image, you can see Hurricane Florence as a system—an organized storm that is separate from other systems or weather patterns.



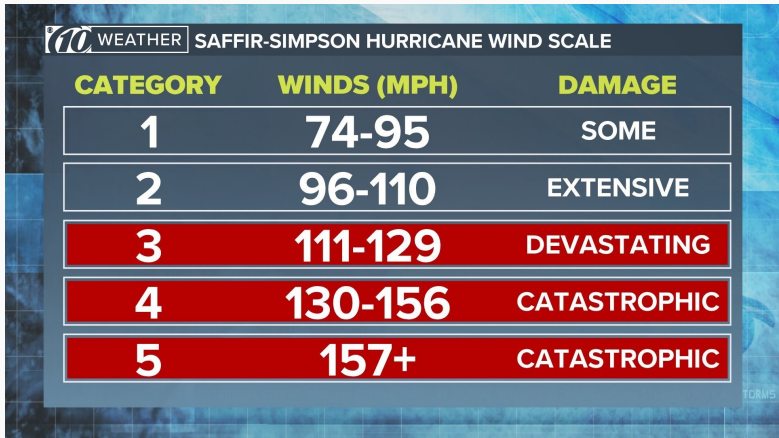
Source: NASA / Goddard Space Flight Center

Central measures of tropical cyclones

One way we track the whole system of a tropical cyclone is to focus on **central measures**. These include measures like the **location** of the center of the storm and the **intensity** at its highest throughout the storm, which is typically near its center.

Central measures of tropical cyclones

One common measure that is used for tropical cyclones as systems is the Saffir-Simpson Hurricane Wind Scale.

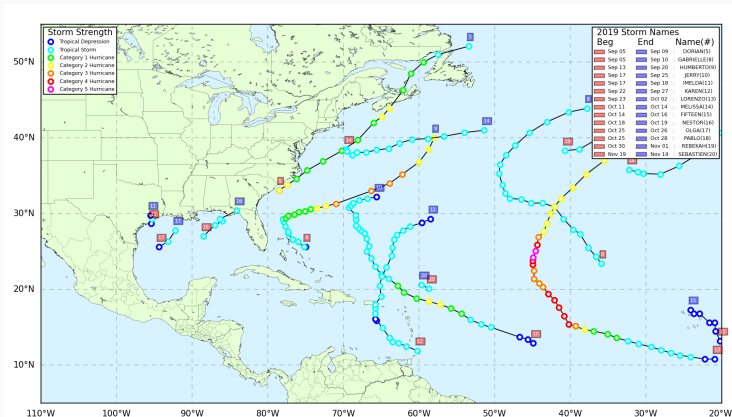
A graphic titled "SAFFIR-SIMPSON HURRICANE WIND SCALE" with a "WEATHER" logo. It features a table with five categories, each with a corresponding wind speed range and damage description. Categories 3, 4, and 5 are highlighted in red. The background is a blue and white grid pattern.

SAFFIR-SIMPSON HURRICANE WIND SCALE		
CATEGORY	WINDS (MPH)	DAMAGE
1	74-95	SOME
2	96-110	EXTENSIVE
3	111-129	DEVASTATING
4	130-156	CATASTROPHIC
5	157+	CATASTROPHIC

Source: WTSP

Storm track data

As we track a storm, we gather these data points at regular times, what are known as the **synoptic times** of 00:00, 06:00, 12:00, and 18:00 UTC (Coordinated Universal Time). For many storms, tracking data will cover **several days**, and in some cases a week or more.



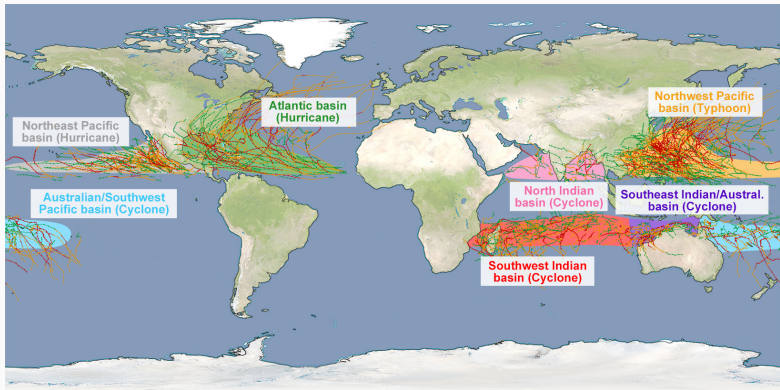
Storm track data are widely available and are going available going back many years. A preliminary version is typically available in real time from the weather service responsible for storm tracking. At a later time, an updated **Best Tracks** version is available.

The tracking data are also available internationally through an effort called the **International Best Track Archive for Climate Stewardship (IBTrACS)**.

To talk about IBTrACS, though, we need to step back and talk about tropical cyclone **basins**.

Tropical cyclone basins

Tropical cyclones occur in different spots in the world, and these spots have been divided into basins:



Source: Deutsche Wetterdienst

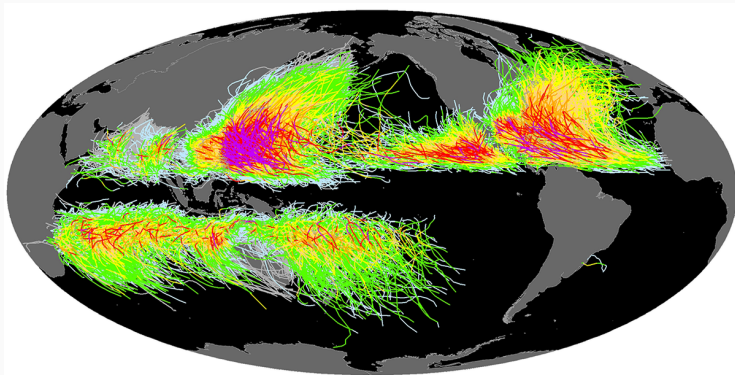
Tropical cyclone basins

Traditionally, different weather services were responsible for tracking tropical cyclones in different basins.

These different weather services sometimes measure things in different ways or with different conventions.

As a result, while you can get tracking data from each responsible group, the resulting data may be hard to compare across study locations.

This issue is addressed by IBTrACS, or the International Best Tracks Archive for Climate Stewardship. IBTrACS provides global, tropical cyclone storm track data, standardized across all the basins.



The International Best Track Archive for Climate Stewardship (IBTrACS) stores global tropical cyclone information.

Saffir-Simpson Hurricane Wind Scale

Intensity Missing	Grey line	Category 1	Yellow line
Tropical Depression	Light blue line	Category 2	Orange line
Tropical Storm	Green line	Category 3	Red line
		Category 4	Dark red line
		Category 5	Purple line

Storm tracks data

If you are studying a single basin, it's fine to use the specific tracking data from the weather service that is in charge of that basin. However, if you are investigating multiple basins, it is best to use the IBTrACS data.

If you do that, it is worth closely reading the IBTrACS documentation, available through its website:

<https://www.ncei.noaa.gov/products/international-best-track-archive>

I also highly recommend this paper:

Knapp et al. 2010. The International Best Track Archive for Climate Stewardship (IBTrACS) unifying tropical cyclone data. *Bulletin of the American Meteorological Society* 91.3: 363-376.

Example

For the US, historical Best Tracks data are available through the `hurricaneexposedata` package in R, with associated functions in the `hurricaneexposure` package. We'll look through that as an example.

Other R packages are available that can help you access the IBTrACS data, including `noaastorms` (<https://github.com/basilesimon/noaastorms/>).

Set-up of example packages

Required set-up

If you'd like to try yourself, there are some set-up steps. First, make sure your computer has R and R studio. If you need them, you can download them from online.

- Install R
- Install RStudio (Recommended but not required)

The data package is stored using a system called `drat`. If you're interested in the details, there's more information here:

CONTRIBUTED RESEARCH ARTICLE

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Hosting Data Packages via `drat`: A Case Study with Hurricane Exposure Data

by G. Brooke Anderson and Dirk Eddebuetel

Abstract Data-only packages offer a way to provide extended functionality for other R users. However, such packages can be large enough to exceed the package size limit (5 megabytes) for the Comprehensive R Archive Network (CRAN). As an alternative, large data packages can be posted to additional repositories beyond CRAN itself in a way that allows smaller code packages on CRAN to access and use the data. The `drat` package facilitates creation and use of such alternative repositories and makes it particularly simple to host them via GitHub. CRAN packages can draw on packages posted to `drat` repositories through the use of the 'Additional_repositories' field in the DESCRIPTION file. This paper describes how R users can create a suite of coordinated packages, in which larger data packages are hosted in an alternative repository created with `drat`, while a smaller code package that interacts with this data is created that can be submitted to CRAN.

Required set-up

All you really need to know, though, is that you need to install the packages in a slightly different way.

- Install `hurricaneexposuredata` (from our personal repo)

```
install.packages("drat") # Only run if you don't have `drat`  
library("drat")
```

```
addRepo("geanders")  
install.packages("hurricaneexposuredata")
```

Required set-up

The other package we'll focus on, `hurricaneexposure` is available through GitHub. You can install that package with the following steps:

- Install `hurricaneexposure` (from GitHub)

```
install.packages("devtools") # If you don't have `devtools`  
library("devtools")  
install_github("geanders/hurricaneexposure")
```

Storm track examples

Example

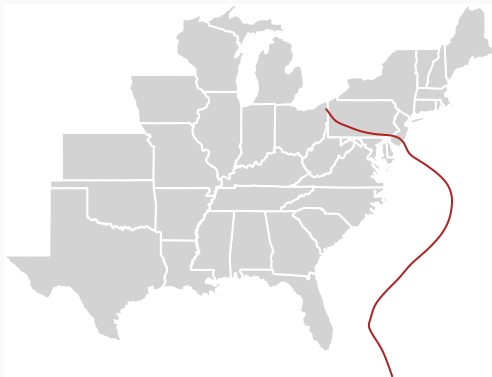
To start, load the two `hurricaneexposure` packages, as well as the `tidyverse` package (for standard data science R tools).

```
library("hurricaneexposedata")  
library("hurricaneexposure")  
library("tidyverse")
```

Example

You can map a storm's track with the `map_tracks` function from `hurricaneexposure`. For example, map the track of Hurricane Sandy with:

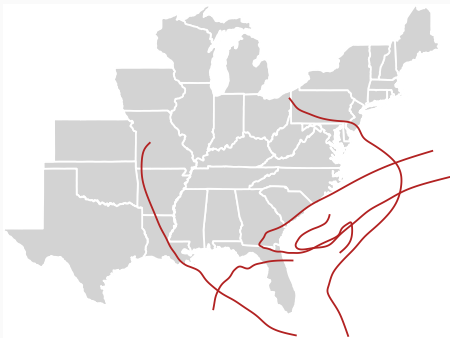
```
map_tracks(storm = "Sandy-2012")
```



Example

You can plot the tracks of multiple storms. For example, you can plot the tracks of all the tracked storms that came near the US in 2012 with:

```
map_tracks(storm = c("Alberto-2012", "Beryl-2012",  
                     "Debby-2012", "Isaac-2012",  
                     "Sandy-2012"))
```



Example

This function is using storm track data that we've included in the `hurricaneexposedata` package, in a dataset called `hurr_tracks`:

```
data("hurr_tracks")  
hurr_tracks %>% sample_n(3)
```

```
## # A tibble: 3 x 6  
##   storm_id    usa_atcf_id date          latitude longitude  wind  
##   <chr>      <chr>      <chr>          <dbl>     <dbl> <dbl>  
## 1 Dolly-2008 AL042008   200807241200    27.2      -98.9    45  
## 2 Fran-1996  AL061996   199608311200    21.1      -61.4    65  
## 3 Arthur-2014 AL012014   201407041200    36.8      -74.4    80
```

Example

You can work directly with this dataset. For example, you can use regular expressions to identify storms that start with “B”:

```
hurr_tracks %>%  
  filter(str_detect(storm_id, "^B")) %>%  
  group_by(storm_id) %>%  
  summarize(n = n(), max_wind = max(wind),  
            start_date = first(date)) %>%  
  slice(1:4)
```

```
## # A tibble: 4 x 4  
##   storm_id      n max_wind start_date  
##   <chr>      <int>    <dbl> <chr>  
## 1 Barry-2001    25      60 200108021200  
## 2 Barry-2007    24      50 200705310000  
## 3 Bertha-1990   37      70 199007241200  
## 4 Bertha-1996   51     100 199607050000
```

By having the data in R, you can leverage many of R's useful other packages:

- `sf` for mapping and working with geospatial data
- `tidyverse` for exploring a plotting data
- `lubridate` for working with dates and times

From central to local measures

From central to local measures

Tracking data can give central measures of the storm, but they don't always give you what you need for epidemiological research.

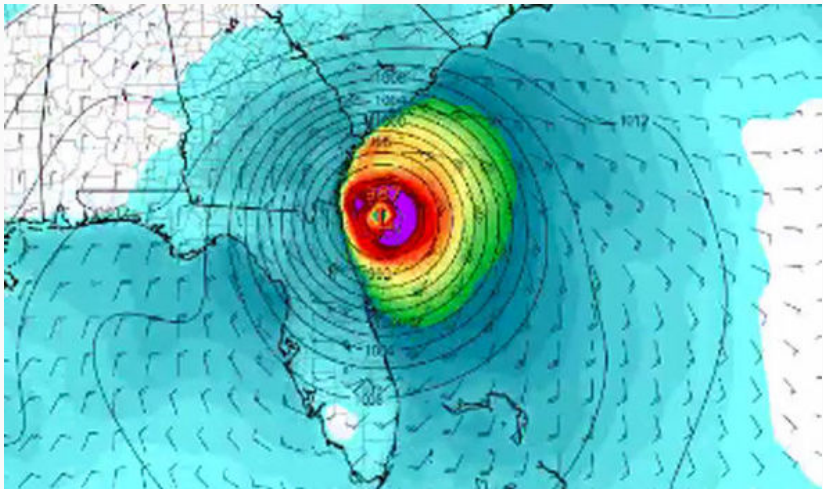
For example, two storms could have the same central storm measures but create very different wind exposures for a location the same distance from each storm track.

From central to local measures

This highlights another important point—many of the people who are affected by a storm do **not** live right along the storm's track. Central measures of the storm won't capture the exposure for anyone who does not live along the storm track.

Wind field models

Instead, it is good to know the **wind field** of the storm. This gives us spatially-defined wind speeds for the storm at a specific time:



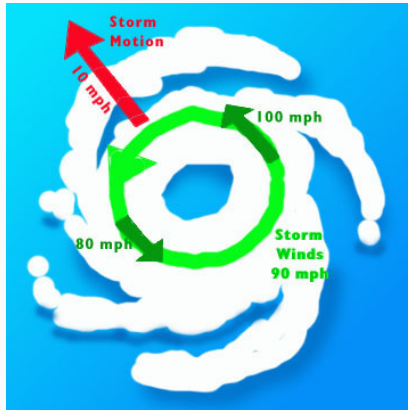
Wind field models are parametric models built to estimate these wind fields.

There are a few you might have heard of, including the Holland model and the Willoughby model.

At heart, each of these gives a smooth fit of storm intensity as a function of the distance from the center of the storm.

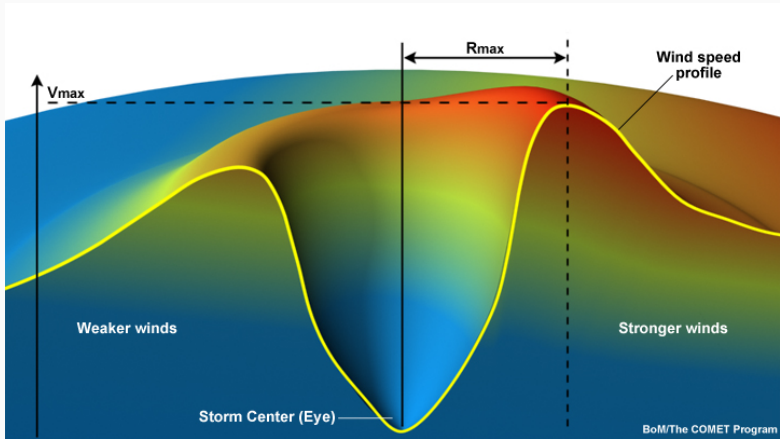
Wind field models

However, tropical cyclones are not symmetrical. This is because the storm is moving in two ways—forward (translational movement and speed) and by rotating (rotational movement and speed). Because storms are moving forward and rotating, wind speeds are higher in certain regions in certain regions of the storm and lower in others.



Wind field models

As a result, when you plot the wind as distance from the storm center, it is asymmetrical.



Source: NOAA

Wind field models

A wind field model like the Holland model takes, as inputs, information from the storm tracks data.

This includes the location of the storm over time, which allows a calculation of the translational (forward) direction and speed of the storm.

It also includes a measure of the central storm intensity, like the maximum sustained winds or the minimum pressure.

To develop these wind field models and tune the parameters that define them, scientists used observational data from historical storms.

A lot of this data, as well as the optimization of the models, aimed to measure the storm right before and at the time of landfall.

Wind field models

These include data from flying airplanes into storms, dropping recording equipment called **dropsondes**.



Wind field models

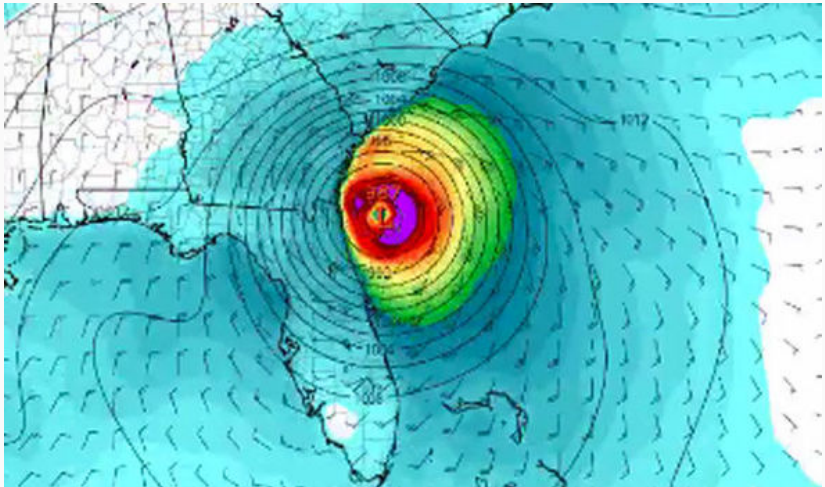
Wind field models can be used to model storm-associated winds not only along the storm track, but also at any location the storm might affect.

These models can therefore be used to generate estimates of storm-associated winds at county or ZIP code centroids, for studies that use aggregated data on health outcomes.

They can also be used to generate wind estimates at residential locations, if data are available at that scale. This would allow you to determine a separate and local exposure estimate for each subject in a study.

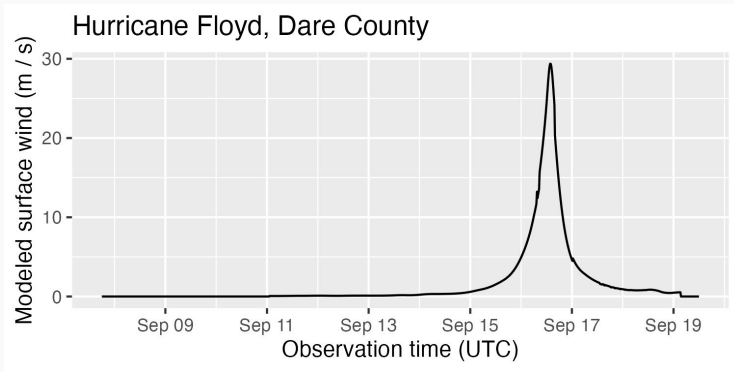
Wind fields

Wind fields usually represent at a single time point during a storm. For example, this wind field for Hurricane Matthew is showing the spatial pattern of its winds right before landfall:



Wind field models

To figure out the peak winds that a storm brought to a location, you can run wind field models regularly over the storm's history.

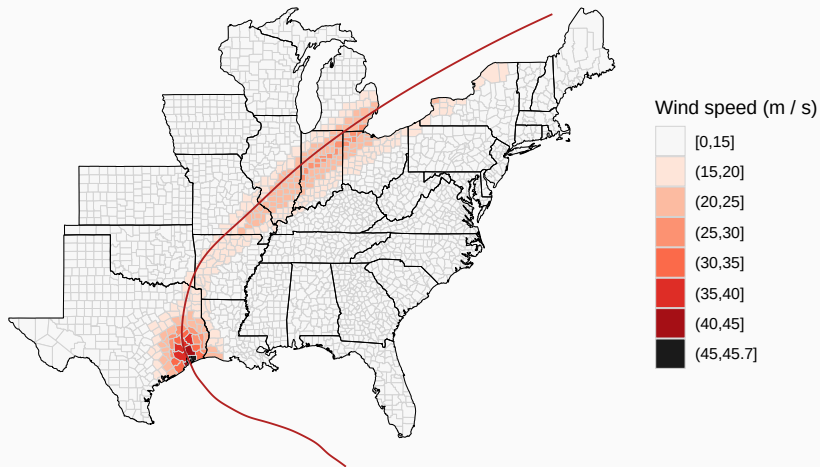


At each location you model, you can create this wind history, then take measures like the peak wind during the storm and the duration of winds of a certain intensity.

Example

Use `map_counties` with `metric = "wind"` to map county-level peak sustained wind from a storm:

```
map_counties(storm = "Ike-2008", metric = "wind")
```



Example

The `county_wind` function from `hurricaneexposure` allows you to determine which of a set of counties had peak sustained wind above a certain threshold, as well as pull information about each location's peak wind and duration of high winds:

```
county_wind(counties = county_centers$fips, wind_limit = 35,  
            start_year = 2008, end_year = 2008) %>%  
  filter(storm_id == "Ike-2008") %>% slice(1:4)
```

```
##      storm_id    fips vmax_sust vmax_gust sust_dur gust_dur usa_atcf_id  
##      <char> <char>    <num>    <num>    <num>    <num>    <char>  
## 1: Ike-2008  48071  45.67278  68.05244      825      1290    AL092008  
## 2: Ike-2008  48167  42.79127  63.75899      840      1290    AL092008  
## 3: Ike-2008  48201  37.14629  55.34797      675      1170    AL092008  
## 4: Ike-2008  48291  43.62377  64.99942      720      1230    AL092008  
##      closest_time_utc storm_dist      local_time closest_date  
##      <char>          <num>          <char>          <char>  
## 1: 2008-09-13 09:00    20.29497 2008-09-13 04:00    2008-09-13  
## 2: 2008-09-13 08:00    21.80563 2008-09-13 03:00    2008-09-13  
## 3: 2008-09-13 10:15    38.91968 2008-09-13 05:15    2008-09-13  
## 4: 2008-09-13 11:00    21.92703 2008-09-13 06:00    2008-09-13
```

A growing number of studies have used wind field models to assess exposure to tropical cyclone–associated winds for epidemiological research and other research on human outcomes.

These include studies of the association between tropical cyclone exposure and:

- mortality (Parks et al., 2023; Blum et al. 2022; Parks et al. 2022, Liu et al. 2023)
- hospitalization (Yan et al. 2021, Parks et al. 2021)
- infectious disease (Huang et al. 2024, Lynch and Shaman, 2023)
- school outcomes (Meltzer et al., 2025)

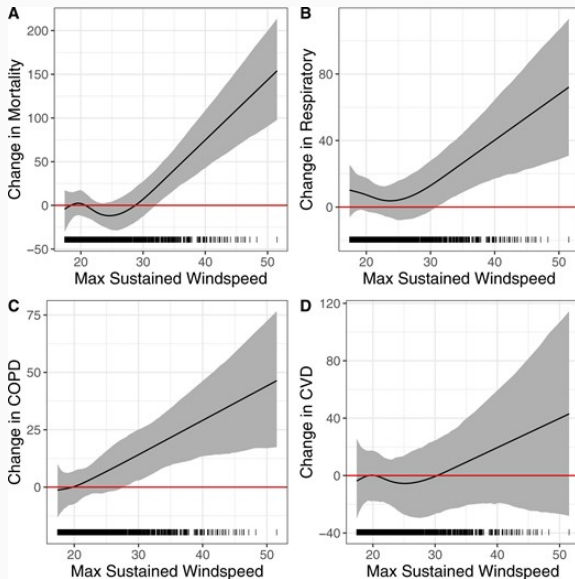
Epidemiologists can use the estimated peak winds from a wind model as either a continuous metric or, in conjunction with a threshold of exposure, to create a binary metric of exposure to the storm.

To date, most studies have looked at exposure as a binary variable.

One exception is a study by Nethery et al. In their analysis, they found evidence of an exposure-response relationship that is very non-linear.

Estimating exposure-response relationships will be important, but is methodologically difficult because of both this non-linearity and the sparsity of data at the highest levels of exposure.

Use in epidemiology



Threshold of exposure

If wind field data are used to classify binary exposure (exposed / unexposed), you must pick a threshold to define exposure.

One common practice in epidemiological studies to date has been to explore results under two or more selections of thresholds of exposure.

Beaufort wind scale

For example, studies might conduct their main analysis using **gale-force wind speed** (34 knots) as the primary threshold, but also include a secondary analysis based on **hurricane-force wind speed** (64 knots).

The names of wind speeds (e.g., gale-force) come from a scale called the Beaufort scale. This scale describes the local conditions created by winds of different speeds.

Beaufort wind scale

Beaufort wind scale for land

The Beaufort wind scale begins at zero with wind speeds of less than one mile per hour

1 Wind speed 1-3mph  Light air: smoke drifts in wind direction	2 Wind speed 4-7mph  Light breeze: wind felt on face, leaves rustle, wind vane moves	3 Wind speed 8-12mph  Gentle breeze: leaves, small twigs in motion, light flags extended	4 Wind speed 13-18mph  Moderate breeze: small branches move, dust, leaves and loose paper raised up	5 Wind speed 19-24mph  Fresh breeze: small trees in leaf begin to sway	6 Wind speed 25-31mph  Strong breeze: large branches move, wind whistles in wires
7 Wind speed 32-38mph  Near gale: whole trees move, resistance felt walking against wind	8 Wind speed 39-46mph  Gale: twigs and small branches break off trees	9 Wind speed 47-54mph  Strong gale: slight structural damage, slates blown off roofs	10 Wind speed 55-63 mph  Whole gale: trees broken, structural damage	11 Wind speed 64-72mph  Storm: widespread damage	12 Wind speed 73mph plus  Hurricane force: violence and destruction

Press Association Graphic

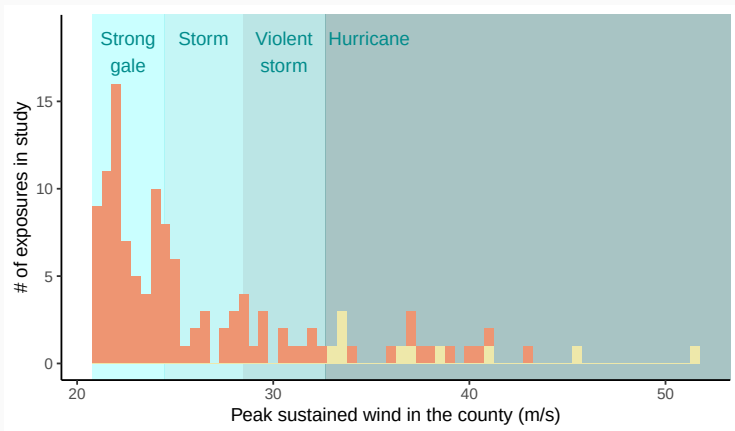
This makes it helpful in determining which threshold might be appropriate based on the hypothesized pathway between exposure and health risk in your study.

For example, if you hypothesize that power outages from the storm are likely to be a driver of risk, it may make sense to consider gale-force as a threshold, since winds at that level can damage trees, which can create significant power outages.

Alternatively, if you are studying psychological trauma related to major damage to personal property, a higher threshold—one associated with damage to buildings—might be more reasonable.

Threshold of exposure

Whichever threshold you select, keep in mind that most of the storm exposures in your study will be right above that threshold. This is because the intensity of wind exposures are highly right-skewed.



Source: Yan et al. 2021

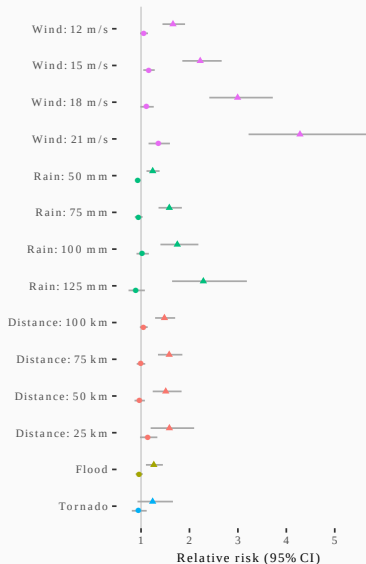
Threshold of exposure

Studies that have considered multiple thresholds of exposure have often found some evidence of an exposure-response relationship, with higher estimated risk when using a higher threshold.

However, the skewed nature of wind exposures means that estimates at a higher threshold often are estimated with much higher variability.

Threshold of exposure

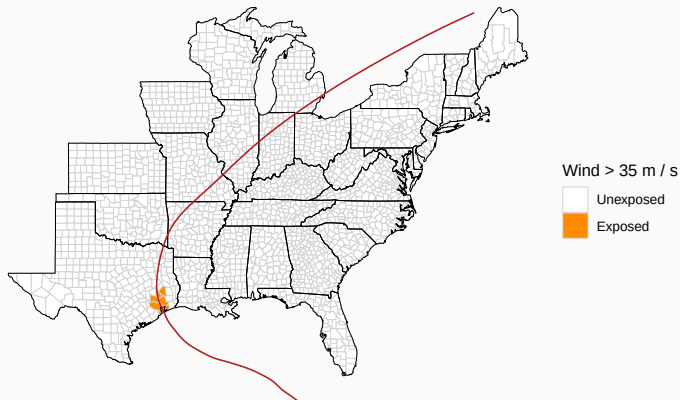
Source: Yan et al. 2021



Example

The `map_wind_exposure` function in `hurricaneexposure` allows you to map all counties with peak sustained wind over a certain threshold (in m/s) for a storm:

```
map_wind_exposure(storm = "Ike-2008", wind_limit = 35)
```



Example

If you want to get values for a certain region (e.g., a state), start by creating a vector with the county FIPs codes for all counties in that region. For example, to get everything from Florida counties, start by running:

```
fl_counties <- county_centers %>%  
  filter(str_sub(fips, 1, 2) == "12") %>%  
  pull("fips")  
fl_counties %>% head()
```

```
## [1] "12001" "12003" "12005" "12007" "12009" "12011"
```

Example

```
county_wind(counties = fl_counties, wind_limit = 45,  
            start_year = 2004, end_year = 2004) %>%  
  select(storm_id, fips, vmax_sust, sust_dur, closest_date)
```

```
##      storm_id    fips vmax_sust sust_dur closest_date  
##      <char> <char>    <num>    <num>    <char>  
## 1: Charley-2004  12015  55.19609      450  2004-08-13  
## 2: Charley-2004  12027  55.07930      480  2004-08-13  
## 3: Charley-2004  12049  45.72336      435  2004-08-13  
## 4: Charley-2004  12071  52.76415      480  2004-08-13  
## 5:  Jeanne-2004  12085  48.76194      885  2004-09-26  
## 6:  Jeanne-2004  12111  50.44674      960  2004-09-26
```

Example

The `hurricaneexposuredata` package provides historical data where the wind field modeling has already been processed, modeling to population-weighted county centers.

We also have a package that can directly run a wind field model, `stormwindmodel`, for cases where you want to model different locations or storms. Computationally, this will take longer, but it provides flexibility beyond the pre-run data in `hurricaneexposuredata`.

There are both advantages and disadvantages to using wind field models for exposure assessment in epidemiological studies.

Advantages:

- Estimates local (rather than storm-wide / centralized) intensity of exposure
- Provides a consistent method to estimate exposure over the study period
- To some degree, provides a consistent method to estimate exposure over the study area

Disadvantages:

- It's a model, and no model is perfect
- Models were optimized using data taken over water or right at the coast. Most people live further inland.
- Models don't perform as well for storms that are traveling over land or for storms that are undergoing extratropical transition
- There are some tropical cyclones with very unusual wind-field patterns

Alternative methods of exposure assessment

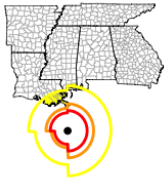
We'll wrap up by talking about two broad topics for alternative exposure assessment:

- Alternative ways to assess exposure to tropical cyclone winds
- Alternative metrics that focus on or incorporate other storm hazards

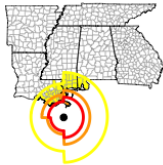
Wind exposure

If you are interested in assessing exposure to the hazard of wind specifically, one alternative to consider are wind radii.

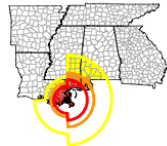
00:00 UTC August 29, 2005



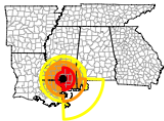
06:00 UTC August 29, 2005



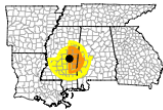
12:00 UTC August 29, 2005



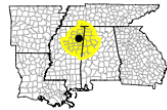
18:00 UTC August 29, 2005



00:00 UTC August 30, 2005



06:00 UTC August 30, 2005



Wind radii

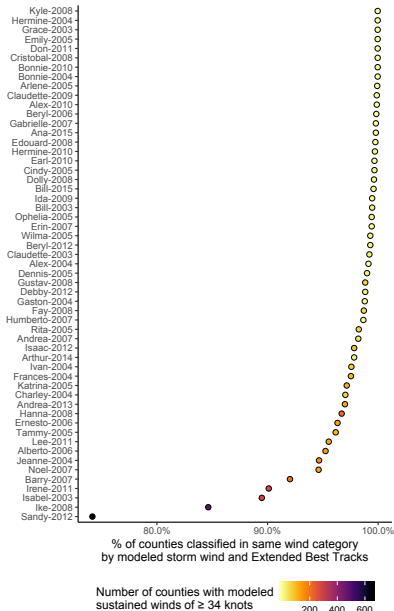
These wind radii are developed through data integration and expert assessment for each synoptic time of each storm. They are therefore storm-specific and capture any unusual characteristics of a specific storm.

For the North Atlantic basin, they are available through the Extended Best Tracks dataset.

There are some disadvantages:

- May not be available for every storm or in every basin
- Not available immediately
- Amount of data they're based on may vary by storm
- Only provide categories of wind speed, not exact values

Comparison of wind data

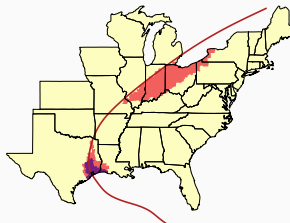
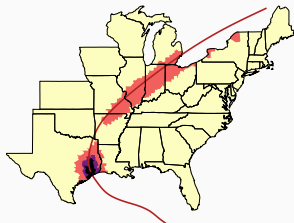


Comparison of wind data

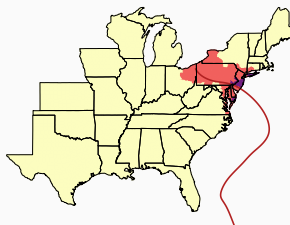
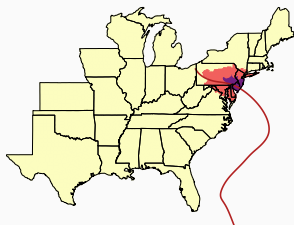
Windspeed 0–33.9 knots 34–49.9 knots 50–63.9 knots 64+ knots

Modeled

Extended Best Tracks



Hurricane Ike
(2008)

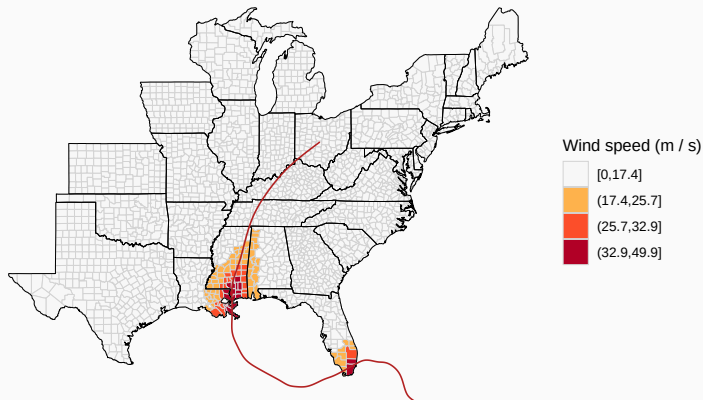


Hurricane Sandy
(2012)

Example

You can change to use the wind radii (instead of the modeled wind) by using the “ext_tracks” option for the wind_source parameter (available for storms 2004):

```
map_counties(storm = "Katrina-2005", metric = "wind",  
             wind_source = "ext_tracks")
```



Other storm hazards

HAZARDS SUMMARY

September 11, 2018 – 8AM

HURRICANE FLORENCE



HAZARD	DETAILS	IMPACTS	THREAT
Storm Surge	S of Cape Fear: 4-6' Cape Fear-Cape Lookout: 6-12' (Neuse & Pamlico Rivers) Cape Lookout-Ocracoke Inlet: 5-8' N of Ocracoke Inlet: 3-5'	Very dangerous inundation amounts are expected along the NC coast Thu-Sat.	Extreme
Inland Flooding	Portions of Eastern NC: 15-20+" Central/Eastern: 6-15" (Up to 30" along track)	Sig. threat to life and property; impassable roads; road wash-outs. Heavy rain will begin Thu and could continue through the weekend. Longer-term river flooding likely, and mountain landslides possible.	Extreme
Damaging Winds	Hurricane-Force winds are likely near the coast Tropical Storm-Force winds likely across much of state	Significant downed trees and widespread/prolonged power outages across the state; significant structure damage possible as Florence could make landfall as a Category 4 hurricane.	Extreme
Tornadoes	A few tornadoes are possible Thu and Fri, especially across eastern NC. Tornadoes in tropical systems are typically short-lived and weak.		Moderate
Marine & Coastal	Life-threatening surf and rip currents will continue for much of the week.		Extreme

Threat Levels:

None

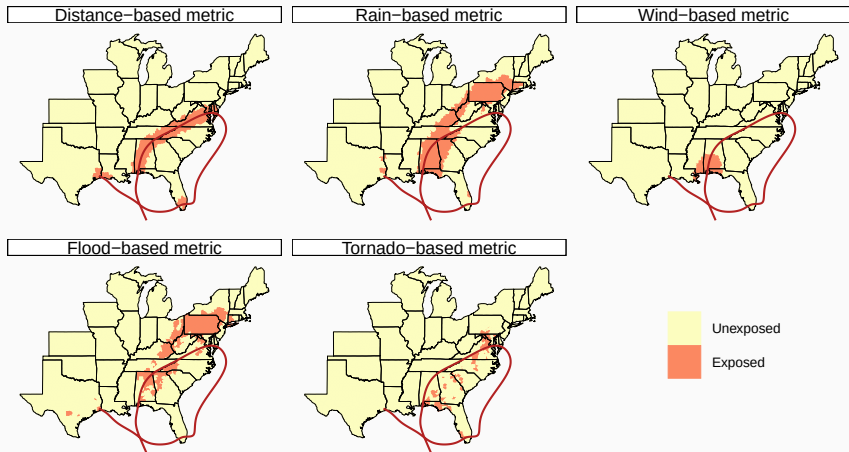
Low

Moderate

High

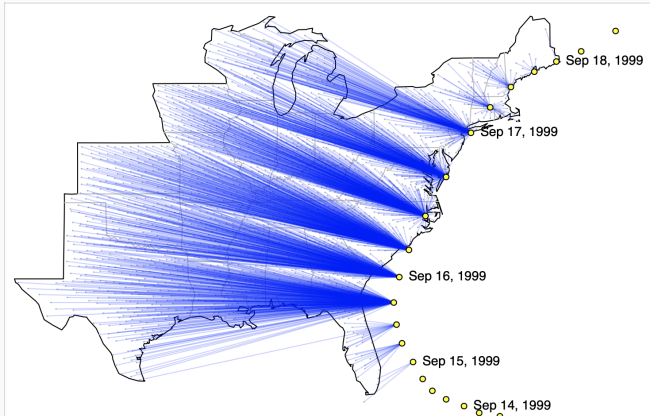
Extreme

Different patterns in hazards



Other storm hazards

For these hazards, you may need to join other datasets by date and location with your study data.



Example

The `county_distance` function from `hurricaneexposure` allows you to determine which of a county were within a certain distance of the storm's track and which date the storm was closest to that county:

```
county_distance(counties = "51700",  
                start_year = 1995, end_year = 2005,  
                dist_limit = 30) %>%  
  select(storm_id, fips, closest_date, storm_dist)
```

```
##      storm_id   fips closest_date storm_dist  
##      <char> <char>      <char>      <num>  
## 1: Bertha-1996 51700   1996-07-13   26.70755  
## 2: Gaston-2004 51700   2004-08-30   15.34549
```

Questions
